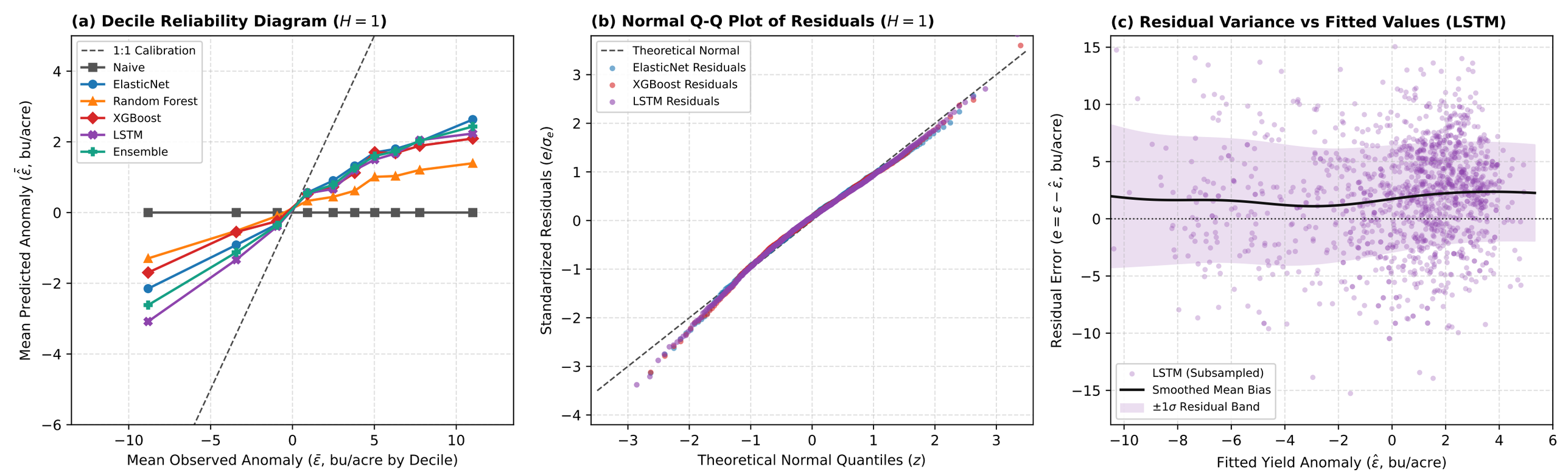

(a) Decile Reliability Diagram ($H = 1$)
1:1 Calibration
Naive
ElasticNet
Random Forest
XGBoost
LSTM
Ensemble
Mean Predicted Anomaly ($\bar{\hat{\varepsilon}}$, bu/acre)
Mean Observed Anomaly ($\bar{\varepsilon}$, bu/acre by Decile)
(b) Normal Q-Q Plot of Residuals ($H = 1$)
Theoretical Normal
ElasticNet Residuals
XGBoost Residuals
LSTM Residuals
Standardized Residuals (e/σ_e)
Theoretical Normal Quantiles (z)
(c) Residual Variance vs Fitted Values (LSTM)
LSTM (Subsampled)
Smoothed Mean Bias
$\pm 1\sigma$ Residual Band
Residual Error ($e = \varepsilon - \hat{\varepsilon}$, bu/acre)
Fitted Yield Anomaly ($\hat{\varepsilon}$, bu/acre)